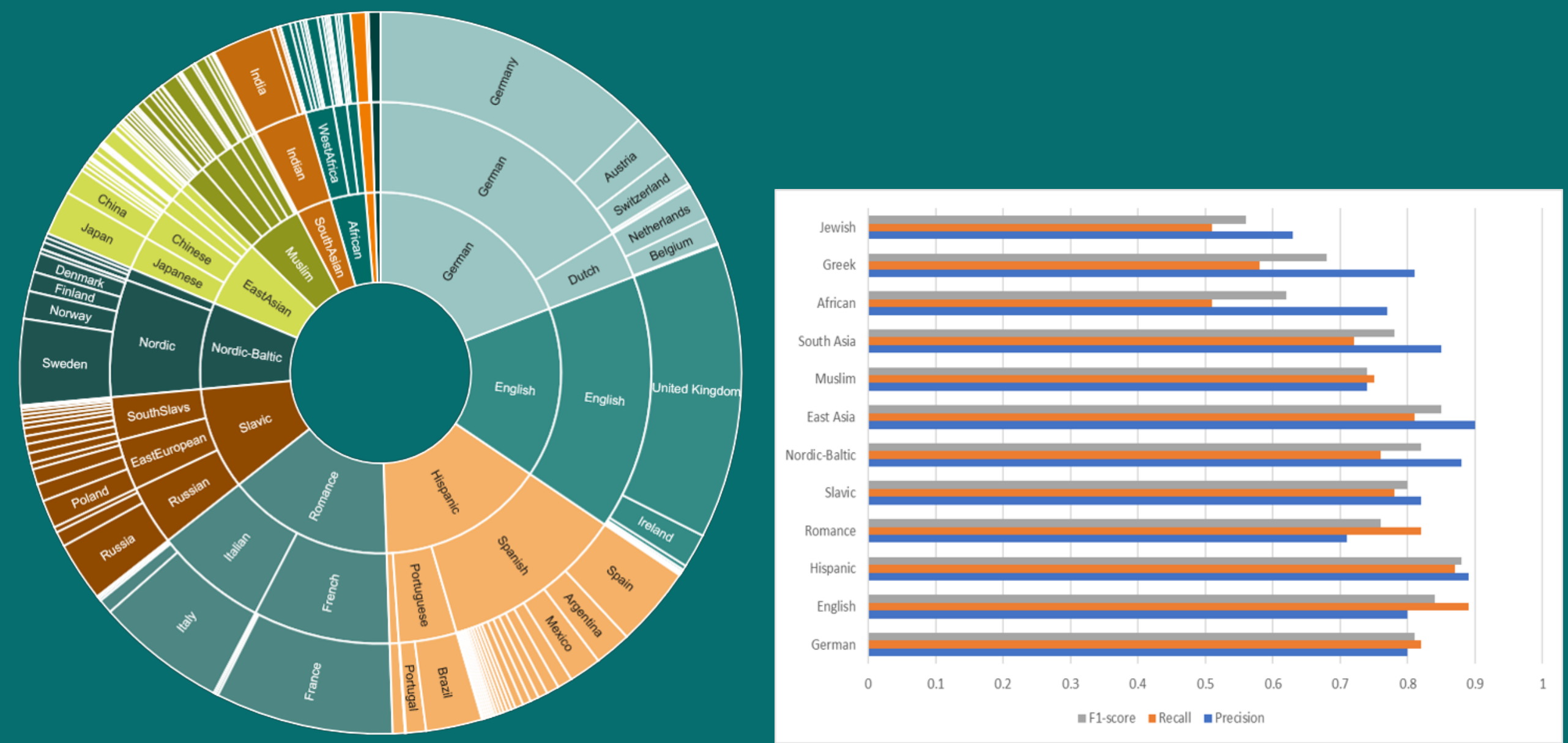


Motivation

- Most digital trace data does not include the nationality of individuals.
- Once this data is used for migration research, it can have a left truncation issue since we are uncertain about the migrant's country of origin.
- Identifying nationality enables a better differentiation between emigration and return migration.
- We detect the nationality with the least available data, full names, using Machine Learning with fasttext and char CNN.
- We use it instead of the country of academic origin in studying the migration of scholars.
- We exclude the main countries of immigration with a high diversity of scholars such as the US, the UK, Canada, and Australia from our test and evaluation steps.
- Once our model shows good performance, we use these data to analyze the composition of scholars' population.



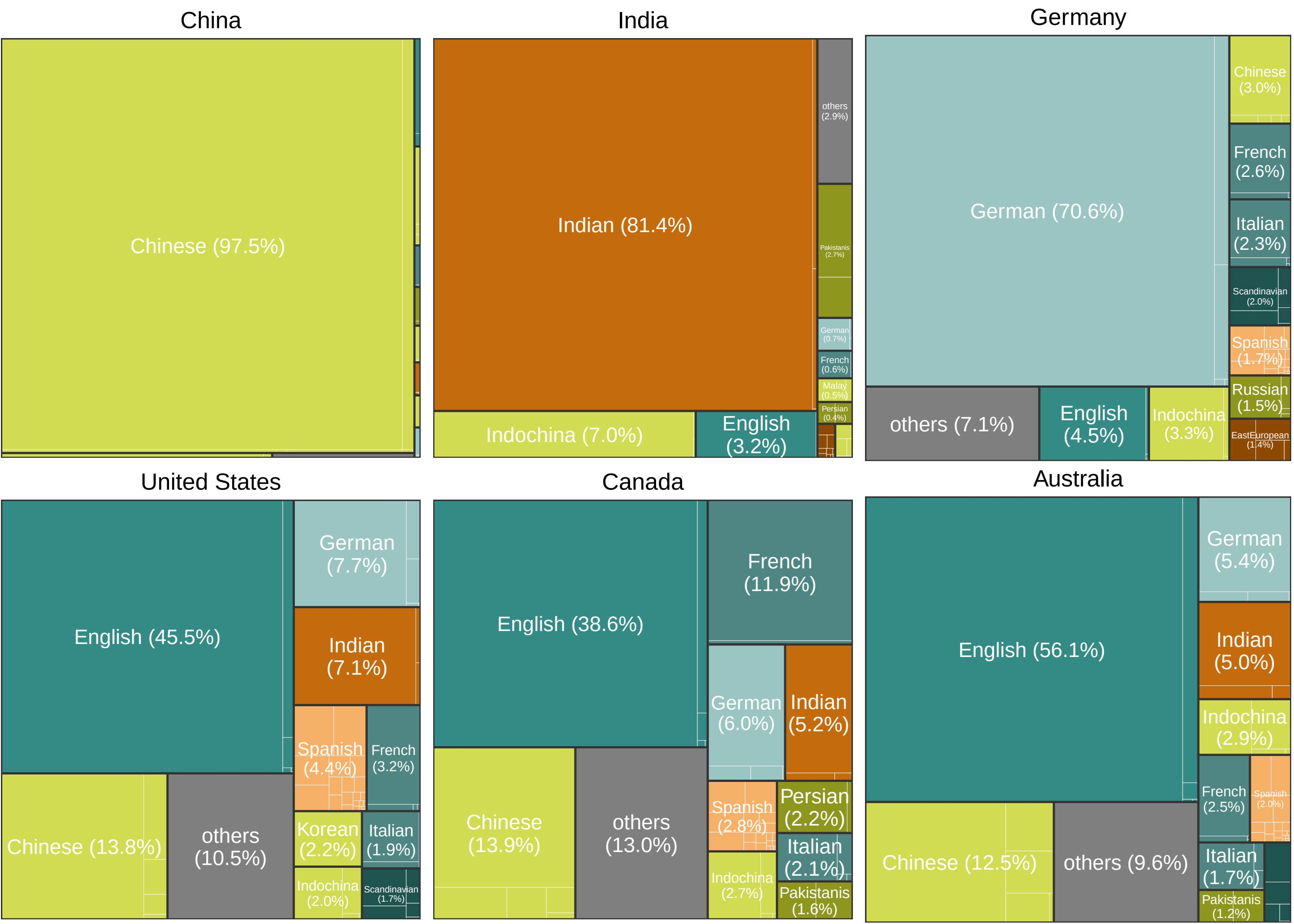
Nested structure of regions and countries of origin based on the name in three levels (left) and performance of our Machine Learning model in level 1 in the most inner circle (right).

Training data used for the Machine Learning and model evaluation data

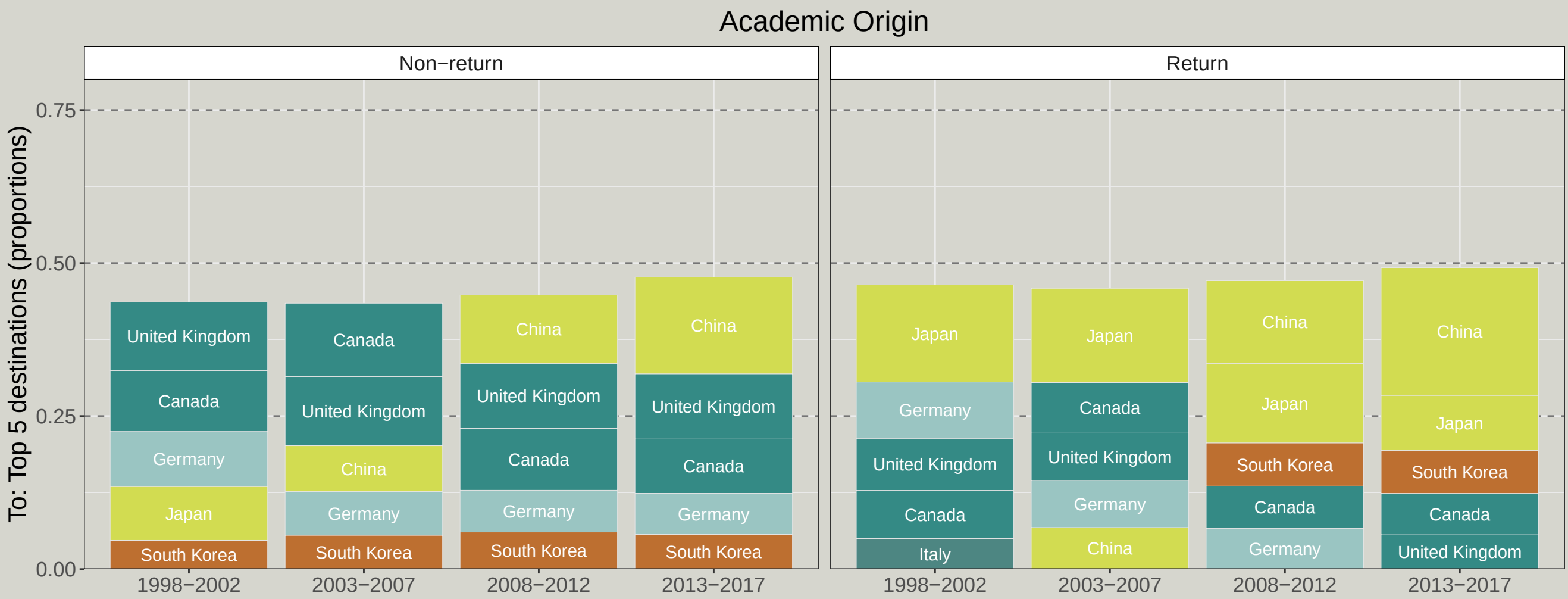
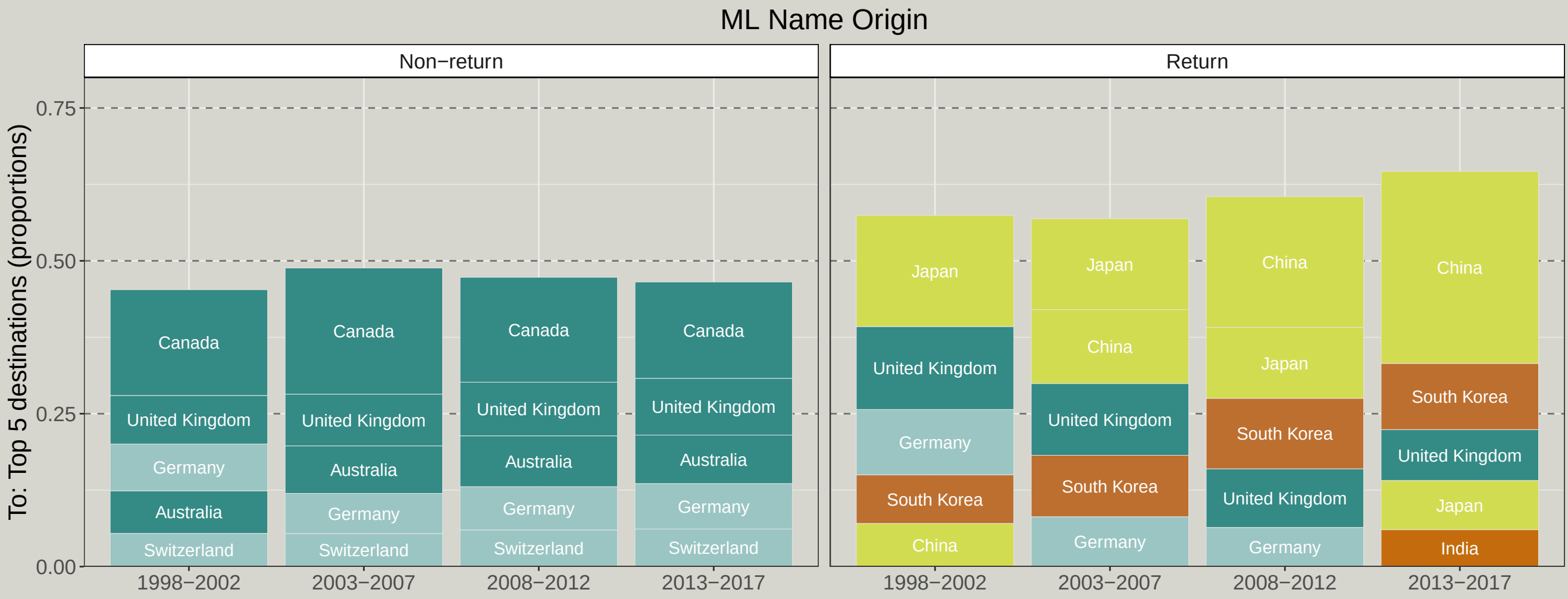
Name	Count
Wikipedia notable people's mobility	2,190,457
Nationality Estimate	696,136
WikiBio (Wikipedia Biography Dataset)	418,978
Wikipedia Person Dataset	355,468
Politicians on Wikipedia	54,877
Pantheon 1.0 dataset	16,071
Wikidata	131,731
Sum	3,874,776
Sum unique	2,634,445

	Level 1		Level 2		Level 3	
	Balanced Accuracy (BA)	Weighted F1	BA	Weighted F1	BA	Weighted F1
IUSSP	72.7	73.4	66.5	66.9	57.1	57.1
Athletes	77.1	77.9	71.8	72.1	58.7	58.5

Population's Composition

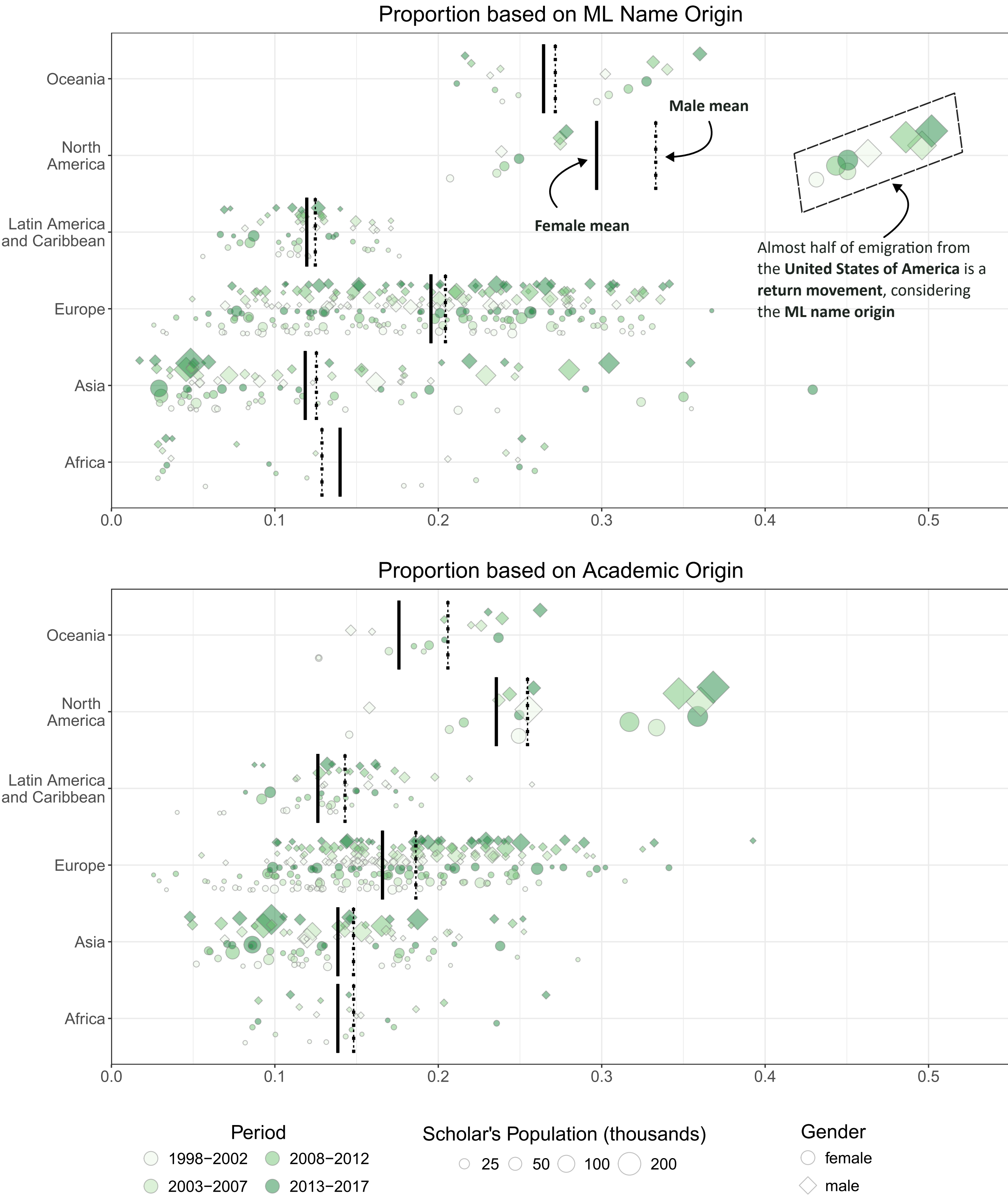


Top 5 destinations for emigrants from the United States



IUSSP members (examples)	Level 1		Level 2		Level 3	
	Reported (R)	Predicted (P)	R	P	R	P
Mohammad Jalal Abbasi-Shavazi	Muslim	Muslim	Persian	Persian	Iran	Iran
Emilio Zagheni	Romance	Romance	Italian	Italian	Italy	Italy
Guy Abel	English	Romance	English	French	UK	France
Mikko Myrskylä	Nordic-Baltic	Nordic-Baltic	Nordic	Nordic	Finland	Finland

Emigration vs. Return Migration



Take Away!

- Summary:** The literature reports high rates of emigration from countries such as the US to China by using the country of first publication as a scholar's academic origin.
- We asked if part of this emigration could be the return of Chinese graduate students and scholars to China and whether assigning nationality using machine learning could help differentiate these cases.
- Results showed that our speculation was correct and identifying a nationality using the name of scholars helps differentiate between emigration and return migration.
- About 50% of emigration from the US was a return migration. 75% of emigration to China is a return migration. In 2013-2017, this increased to 87%. It was 26% and 43% using academic origin, respectively.
- Limitations:** Our ML model works best for names from some countries such as China, Japan, and South Korea. Training data is over-represented by Western names. ML model's performance drops for Spanish and Southern American names that are hard to identify the country of origin.
- Next steps:** We will calculate reliability measures for different migration corridors where our model's results are the most reliable.